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Fin-STance: A Novel Deep Learning-Based Multi-Task Model for Detecting Financial Stance and Sentiment

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Abstract—Stance detection has been gaining popularity in text mining and information retrieval-based research. Recognition of sentiment also plays an important role in the analysis of text whether the underline stances are in *favor* or *against* for a particular domain. However, there is extensive research in this direction, but considering its impact on the financial text has not been explored yet. This paper presents a new deep learning based multi-task approach for financial stance detection and sentiment detection. A new model called **Fin-STance** is introduced which performs two-classification tasks as stance detection and sentiment detection based on the financial data. It constitutes of an input, embedding, followed by shared and task-specific layers. The empirical evaluation of this study is conducted on a newly created dataset and it shows impressive results. Also, this study receives better results with relevant compared methods.

Index Terms—online social media, stance detection, financial stance detection, financial sentiment analysis, information retrieval

I. INTRODUCTION

The Internet has evolved into a crucial platform to generate and retrieve a vast volume of user-generated information [1]. These information is broadly found with unstructured semantics and structure [2]. Interestingly, it has now a beneficial source for computation linguistics researchers world-wide [3]. Researchers perform extensive research on numerous text mining applications, including sentiment analysis and opinion mining [4]. It is done via extraction of important information from these generated online content [5]. Stance detection is considered as one of the sub-tasks of opinion mining and it is likewise a sentiment analysis problem [6]. It analyses the textual portion on a particular topic and its associated comments to determine whether it is in ‘*favor*’, ‘*against*’, or ‘*neither*’ to explore the sentiment in terms of polarity as *positive*, *negative*, or *neutral* for different opinions [7]. Stance detection is applied to several applications like fake news detection, analyzing social media data, argument search, etc [8].

The role of stance has also now become essential in textual data to understand the authenticity of financial information [9]. The understanding of crucial information through opinions and sentiments like financial text is paramount important

to fathom the true impact of financial opinions [10]. It is achieved by taking stance-based decisions that provide invaluable insights and give direction in decision-making. Therefore, it works as actionable information for analysts, investors, and policymakers in financial organizations which can leverage stance-based informed decisions with confidence to have a strategic edge over their rivals, especially in the current macroeconomic era [11]. Also, it is advantageous for financial debates, determining the stock market trends, verification of financial fake news, etc. Admitting such facts, the exploration of financial stance detection in line with sentiment is a significant problem.

A. Our Contributions

In this study, we propose a deep learning-based multi-task approach for two-classification tasks (stance detection and sentiment detection) in financial texts. To this end, a new model called **Fin-STance** is introduced which takes a financial topic and its respective argument as textual inputs which are passed to the contextual embedding layers. Thereafter, the resultant outcome of the embedding layer is passed to the shared layers, which operate as a parallel network, wherein convolutions neural network (CNN) [12] extracts low-level financial stance and sentiment related syntactic and semantic information, bi-directional gated recurrent network (BiGRU) [13] is responsible for capturing latent contextual sequences, and attention layer is used to receive important information from relevant financial stance and sentiment-based keywords from two parallel networks. Thereafter, a common fusion layer is used which combines context vectors of two parallel attention layers. The resultant combined context vector is passed to a common dense layer. Thereafter, the task-specific layers receive dense layer outcomes. It also operates as two parallel networks. One of the networks performs stance detection, whereas another one is used for sentiment detection. Both parallel networks constitute a high-level attention layer to receive more detailed contextual information followed by dense and output layers separately for stance detection and sentiment detection for financial texts.

The key contributions of this study are given below:

- Explored a multi-task approach to address finance-based stance and sentiment detection problem.
- Developed a new *Fin-STance* model to detect financial stance and sentiment detection.
- Formation of a new dataset for the proposed *Fin-STance* model.
- Conducted the experimental evaluation of the proposed model and compared with relevant baselines methods.

The rest of this study is presented as follows: Section II presents the existing literature. The proposed multi-task approach is presented in Section III. It includes the problem description, detail about the newly created dataset followed by pre-processing steps, and the proposed *Fin-STance* model. Section IV presents the experimental setup and results. Lastly, Section V concludes this study and highlights future works.

II. RELATED WORK

The section presents the existing studies for stance detection. In [14], authors proposed n -grams-based dictionary and reliant features for stance detection. In [15], authors considered support vector machine to perform stance detection to achieve a higher accuracy. In [16], authors considered topics in textual data for generative standpoint classifier. It followed an adaptive method to determine parameter values to detect stance. In [17], authors considered emotion and lexicons to create a heterogeneous graph based on semantic-emotion. It fed into a graph convolutional networks (GCN) to capture semantic connections across emotion tags and words to detect stance. In [18], authors captured contextual words via target-adaptive pragmatic dependency graphs and GCN blocks for each tweet for stance detection. In [19], authors predicted stance in bipolar affinities on election-related tweets. In this line, they proposed a fast stance detection framework. In [20], authors detected multi-target stance detection via dynamic memory-augmented network. They used external memory for this purpose in textual data.

In [21], authors introduced a new model *CompGCN*, which is based the commonsense knowledge. It exploited semantic and structural-level information via knowledge graph with an aim to improve the reasoning and generalization characteristics to detect stance. In [22], authors collected external resources to detect stance. They participated in *SemEval-2016*, wherein position detection of CNN is discussed. In [23], authors considered LSTM and GRU in multi-label stance detection and found BiLSTM performed better. In [24], authors considered encoder-decoder framework based on attention network and it performed better as compared to the multi-target stance detection. In [25], authors considered stance and sentiment in their work. They highlighted its effectiveness via experiments, but they used only CNN with one filter only. Further, in [11], authors considered CNN with two-channels of CNN, followed by GRU, and pooling layers in parallel. They have used a concatenated layer which gives a combined vector from parallel networks. They found better results and obtained more semantic information for stance detection.

A. Current Status and Limitations

Stance detection and sentiment is a well-studied problem by researchers, but financial stance detection with sentiment in parallel is not explored. Considering these facts, the detection of finance-based stance and associated sentiment tasks are worth research problems and need of the hour. Considering this, to the best of our knowledge, the proposed multi-task approach for detecting financial stance and sentiment is a new research direction.

III. PROPOSED MULTI-TASK APPROACH

This section presents the proposed approach which includes the problem description, newly collected dataset and its pre-processing steps followed by brief detail of all the layers of the proposed *Fin-STance* model.

A. Problem Description

This study refers to a multi-task approach, wherein two-tasks are taken into consideration. One of the tasks is financial stance detection whereas another relevant task is sentiment detection. Both tasks refer to two-class problems. It infers that in the stance detection task, a finance-based textual input data is detected as either *financial favorable stance* (FFS) or *financial against stance* (FAS). Also, the same input is classified as either *financial positive sentiment* (FPS) or *financial negative sentiment* (FNS) for sentiment detection task.

B. Dataset

In this study, we have created a new dataset called *FST* based on financial stance topics. We have collected finance-based stance text from four sources (*DebateWise*¹, *ProCon*², *idebate*³, and *Kialo*⁴) using *Selenium*⁵ and *Scrapy*⁶ Web scrapping tools. We have collected *topics* with its *descriptions* followed by *title*, *argument* and finally stance-based label as either '*favor*' or '*against*'. Table I presents the final statistics of the newly collected *FST* dataset from four different sources.

TABLE I
FINAL STATISTICS OF THE NEW *FST* DATASET TAKEN FROM DIFFERENT SOURCES

Source	Favor	Against	Total
DebateWise	150	150	300
ProCon	145	152	297
idebate	838	838	1676
Kialo	18949	17529	36478
Total	20082	18669	38751

¹<https://debatewise.org/>

²<https://www.procon.org/>

³<https://idebate.net/>

⁴<https://www.kialo.com/>

⁵<https://www.selenium.dev/>

⁶<https://scrapy.org/>

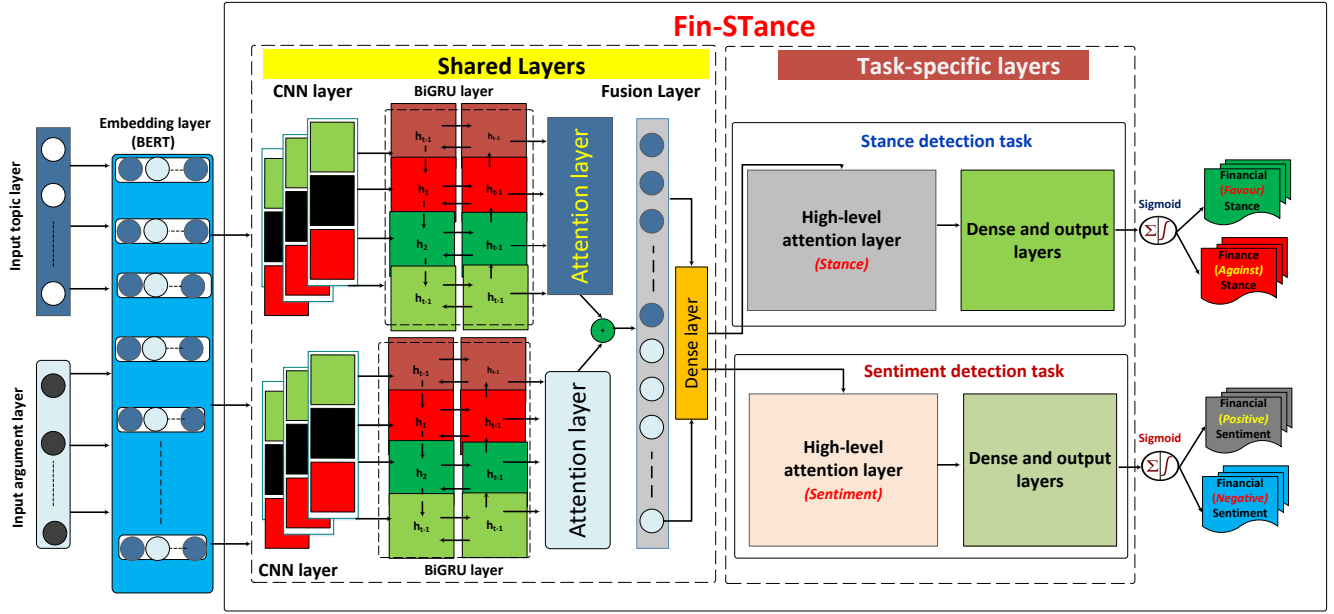


Fig. 1. Work-flow of the proposed Fin-STance model

Besides stance labelling task in the new FST dataset, we calculate the sentiment of each data points via FST dataset using SiBERT [26] (*aka*, Sentiment in English), a pre-trained model based on RoBERTa to perform binary sentiment classification. It is used with an aim to label each data point as either *positive* or *negative* for sentiment detection.

C. Data Pre-processing

We have performed numerous pre-processing tasks on the collected data as mentioned in table I to obtain a refined and pre-processed FST dataset which can be used in this study. Hence, we have removed numbers, comma, exclamation, quotation, punctuation, full-stops, and convert the input raw data into its lower-case form.

D. Fin-STance Model

This section presents the proposed Fin-STance model, as given in figure 1. It gives the complete description of all the layers of the two-tasks for classification (*stance detection* and *sentiment detection*)

1) *Input Layer*: The input layer receives finance-based pre-processed topic and argument as input texts in parallel. Each input word is tokenized in both topic and argument texts to assigned a numeric value as an index. Finally, both input files turn into two dictionaries and converted into two numeric vectors as v_t and v_c which match with the index value of their respective dictionaries. Thereafter, two numeric vectors are converted to a fixed-padding length f_t and f_c of 128 separately to maintain the same length of each input topic and argument vectors. As a result, two padded vectors p_t and p_c of fixed length size are forwarded to the succeeding layer.

2) *Embedding Layer*: In this study, we have used BERT [27], a popular pre-trained language model. It receives two padded input vectors p_t and p_c as input for the BERT embedding layer. It retrieves relevant and enriches contextual financial stance and sentiment-related information in both directions via 768-dimensional word vector representation using Transformer. Finally, the encoded representation using BERT is forwarded to the shared layers.

3) *Shared Layers*: In this layer, CNN, BiGRU, and attention layers are used as parallel networks. The purpose of the shared layers in the proposed multi-task approach is to capture low-level semantic, syntactic, and contextual financial stance and sentiment-related information in parallel mode of input based on the topic and its corresponding arguments.

- **CNN Layer**: In this study, CNN is mainly used in the proposed Fin-STance model to extract relevant finance-based semantic and syntactic features for stance and sentiment via convolutional and pooling operations. The proposed Fin-STance model takes a one-dimensional convolution layer with total 256 filters and a window size of 3. All filters perform convolutional operation. The pooling layer is taken to retrieve the maximum relevant features. It extracts features of numerous size sequences. Consequently, CNN generates feature sequences for all 256 filters. Equation 1 shows n^{th} feature sequence f_n extracts from the convolutional operation, where w shows weight and b indicates bias, and $r(\cdot)$ refers to ReLU, a non-linear activation function.

$$f_n = r(w \cdot x_t + b) \quad (1)$$

- **BiGRU Layer:** BiGRU, a kind of recurrent neural network is operational in opposite directions with two gates intending to produce stance and sentiment-related contextual and latent financial semantic feature sequences with the total number of 64 neurons. It obtains latent semantic representation of stance and sentiments by utilizing financial tokens as forward and backward sequences in opposite directions, as mentioned in equations 2 and 3. Thereafter, equations 4 give a combined hidden representation of GRU.

$$\overrightarrow{gru}_f = \overrightarrow{GRU}(L_{m_t}), n \in \quad (2)$$

$$\overleftarrow{gru}_b = \overleftarrow{GRU}(L_{m_t}), n \in \quad (3)$$

$$gru_i = [\overleftarrow{gru}_b, \overrightarrow{gru}_f] \quad (4)$$

- **Attention Layer:** Attention layer highlights important financial stance and associated sentiment-related keyword-based features. It reduces the presence of normal input tokens. As a result, it gives two resultant context vectors as an output for two attention layers (i.e., one for the financial stance and another for the financial sentiment).
- **Fusion Layer:** This layer concatenate two context vectors generated from two attention layers. Consequently, a comprehensive context vector is generated which consists of both stance and sentiment based important information from relevant keywords.
- **Dense Layer:** This layer receives the output of the fusion layer. It gives the outcome to the task-specific layers, wherein finance-based stance and sentiment detection tasks are employed.

4) *Task-Specific Layers:* The task-specific layers receive the outcome from the shared layers as a resultant vector from the attention layer. The main goal of the task-specific layers is to capture features at a higher level of abstraction which are more specific to finance-based stance and sentiment detection tasks. The task-specific layers operate in parallel for both tasks. Each task in task-specific layers constitute a high-level attention layer followed by dense and output layers.

- **High-Level Attention Layer** Two high-level attention layers are used in task-specific layers which receive dense layer outcomes used in the shared layers for both financial stance and sentiment detection tasks. The purpose of this high-level attention layer is to obtain more fine-grained contextual information for financial stance and sentiment. As a result, it gives two high-level context vectors as f_s and f_t for stance detection and sentiment detection tasks, respectively.
- **Dense and Output Layers:** Two fully dense layers are used in parallel to give financial stance and financial sentiment-based features set from two high-level context vectors f_s and f_t , respectively. Each dense layer is divisible into two classes, wherein one is used for stance

detection and another is used for sentiment detection. *Sigmoid* and *binary cross-entropy* are used separately for the final classification task for stance detection and sentiment detection.

TABLE II
EXPERIMENTAL AND HYPERPARAMETER SETTINGS

Experimental settings		Hyperparameter settings	
Environment	Configuration	Parameters	Value
Operating system	Ubuntu-20.04	Embedding dimension (BERT)	768
Processor	Intel Haswell	CNN Window size	3
Memory	64 GB	BiGRU — #neurons	64
Scripting language	Python 3.9	Spatial dropout	0.2
Library	Keras 2.2.4	Padding sequences	128
GPU	NVIDIA Tesla	CNN — #filters	256
Optimizer	Adam	Epoch (early stopping)	30

IV. EXPERIMENTAL SETUP AND RESULTS

This section demonstrates the experimental setup and evaluation results. We have divided final dataset as shown in table I into training, testing, and validation as 70%, 20%, and 10%, respectively. Table II presents the experimental and hyperparameter settings.

A. Evaluation Results

We have considered four evaluation metrics – *precision*, *recall*, *f-score*, and *accuracy* for experimental results evaluation. It is observed from table III that our Fin-STance model shows impressive classification results for both finance-based stance and sentiment detection tasks. It receives *f-score* values of 0.91 and 0.88 for financial stance detection and sentiment detection, respectively. Also, table IV shows that our Fin-STance model performs better in terms of training versus test accuracy values for both tasks. The proposed model receives 0.90 and 0.88 training accuracy and test accuracy, respectively for task-A (financial stance detection). Likewise, it receives 0.85 and 0.84 training accuracy and test accuracy, respectively for task-B (financial sentiment detection).

B. Comparison with Existing Methods

This section presents the comparison of the Fin-STance model with existing methods. However, there is no study exists for comparison with our work, but despite that, a relevant existing literature from Li et al. [11] work is taken for comparison with our study.

It performs 35.82% and 31.34% better as compared to Li et al. [11] work for financial stance detection and financial sentiment detection tasks, respectively in terms of *f-score*. Moreover, we have also compared results with several baselines methods. Table III shows that the proposed Fin-STance model receives better results for both finance-based task-A (stance detection) and task-B (sentiment detection) as compared to the [11] work. BiGRU receives better results across all neural network baseline methods in terms of *precision*, *recall*, and *f-score*. The proposed Fin-STance model performs 22.97% significantly better as

TABLE III

PERFORMANCE EVALUATION RESULTS ON NEWLY CREATED FST DATASET ON THE **PROPOSED Fin-STance MODEL** FOR FINANCE-BASED STANCE DETECTION (TASK-A) AND SENTIMENT DETECTION (TASK-B)

Methods ↓	Precision	Recall	F-score
Proposed Fin-STance model (Task-A)	0.88	0.94	0.91
Proposed Fin-STance model (Task-B)	0.86	0.90	0.88
Li et al. [11]	0.64	0.71	0.67
CNN	0.68	0.76	0.72
GRU	0.65	0.64	0.64
LSTM	0.61	0.60	0.60
BiGRU	0.70	0.77	0.74
BiLSTM	0.65	0.69	0.67

TABLE IV

TRAIN ACCURACY VERSUS TEST ACCURACY ON NEWLY CREATED FST DATASET ON THE **PROPOSED Fin-STance MODEL** FOR FINANCE-BASED STANCE DETECTION (TASK-A) AND SENTIMENT DETECTION (TASK-B)

Methods ↓	Train Accuracy	Test Accuracy
Proposed Fin-STance model (Task-A)	0.90	0.88
Proposed Fin-STance model (Task-B)	0.85	0.84
Li et al. [11]	0.72	0.72
CNN	0.73	0.70
GRU	0.70	0.64
LSTM	0.62	0.68
BiGRU	0.66	0.69
BiLSTM	0.65	0.67

compared to BiGRU for financial stance detection in terms of *f-score*. Likewise, it performs 18.91% better than BiGRU for financial sentiment detection task in terms of *f-score*.

Table IV shows that the proposed Fin-STance model outperforms both state-of-the-art and baseline methods for both task-A and task-B in terms of training and testing accuracy values. It performs 25.00% in terms of training accuracy and 22.22% better in terms of testing accuracy values as compared to Li et al. [11] work for financial stance detection task. Similarly, it performs 18.05% and 16.66% better in terms of training accuracy and testing accuracy values, respectively as compared to Li et al. [11] work for financial sentiment detection task. CNN performs better in terms of training and testing accuracy values. The proposed Fin-STance model performs 23.28% and 25.71% better in terms of training accuracy and testing accuracy values, respectively as compared to CNN for financial stance detection task. Likewise, it performs 16.43% and 20.00% better in terms of training accuracy and testing accuracy values, respectively as compared to CNN for financial sentiment detection task.

V. CONCLUSION AND FUTURE WORKS

In this study, we have proposed a deep learning-based multi-task approach to accomplish two tasks (financial stance detection and financial sentiment detection). A new model called Fin-STance is introduced for this purpose. It is consisted of an input, embedding, followed by shared and

task-specific layers to perform classification of two tasks (stance detection and sentiment detection) on a financial input text. A new labeled dataset has formed in this study. The proposed Fin-STance model has remarkably outperformed as compared with baseline methods. The inclusion of multi-modal settings and multi-lingual data could be an interesting research directions for this study.

REFERENCES

- [1] Khan, S., Fazil, M., Sejwal, V. K., Alshara, M. A., Alotaibi, R. M., Kamal, A., and Baig, A. R.: BiCHAT: BiLSTM with Deep CNN and Hierarchical Attention for Hate Speech Detection. Journal of King Saud University-Computer and Information Sciences, 34(7), pp. 4335–4344 (2022)
- [2] Kamal, A., Abulaish M.: Self-deprecating humor detection: A machine learning approach. In Proceedings of the Computational Linguistics: 16th International Conference of the Pacific Association for Computational Linguistics, pp. 483-494. Springer Singapore, Hanoi, Vietnam (2020)
- [3] Kamal A., Abulaish M.: An LSTM-based Deep Learning Approach for Detecting Self-deprecating Sarcasm in Textual Data. In Proceedings of the 16th International Conference on Natural Language Processing (ICON), pp. 201–210. ACL, IIIT Hyderabad, India (2019)
- [4] Abulaish, M., and Kamal, A.: Self-Deprecating Sarcasm Detection: An Amalgamation of Rule-Based and Machine Learning Approach. In Proceedings of the IEEE/WIC/ACM International Conference on Web Intelligence, pp. 574–579. IEEE, Santiago, Chile (2018)
- [5] Kamal, A., Anwar, T., Sejwal, V.K. and Fazil, M.: BiCapsHate: Attention to the Linguistic Context of Hate via Bidirectional Capsules and Hatebase. IEEE Transactions on Computational Social Systems (TCSS), pp.134–141 (2023)

- [6] Kamal, A., and Abulaish, M.: Cat-bigru: Convolution and Attention with Bi-directional Gated Recurrent Unit for Self-Deprecating Sarcasm Detection. *Cognitive Computation (COGN)*, pp.1–19 (2021)
- [7] Khan, S., Kamal, A., Fazil, M., Alshara, M. A., Sejwal, V. K., Alotaibi, R. M., Baig, A.R., and Alqahtani, S.: HCovBi-caps: Hate Speech Detection using Convolutional and Bi-directional Gated Recurrent Unit with Capsule Network. *IEEE Access*, 10, pp. 7881–7894 (2022)
- [8] Abulaish, M., Kamal, A., and Zaki, M. J.: A Survey of Figurative Language and its Computational Detection in Online Social Networks. *ACM Transactions on the Web (TWEB)*, 14(1), pp. 1–52 (2020)
- [9] Mohankumar, P., Kamal, A., Singh, V.K. and Satish, A.: Financial Fake News Detection via Context-Aware Embedding and Sequential Representation using Cross-Joint Networks. In *Proceedings of the 15th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pp. 780–784. IEEE, Bengaluru, India (2023)
- [10] Kamal A, Abulaish M.: IMFinE: An Integrated BERT-CNN-BiGRU Model for Mental Health Detection in Financial Context on Textual Data. In *Proceedings of the 19th International Conference on Natural Language Processing (ICON)*, pp. 139–148. ACL, IIIT Delhi, India (2022)
- [11] Li, W., Xu, Y. and Wang, G.: Stance detection of microblog text based on two-channel CNN-GRU fusion network. *IEEE Access*, 7, pp.145944–145952 (2019).
- [12] Kim Y:Convolutional Neural Networks for Sentence Classification, In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp.1746–1751, (2014)
- [13] Chung, J., Gulcehre, C., Cho, K. and Bengio, Y.: Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555*.
- [14] Anand, P., Walker, M., Abbott, R., Tree, J.E.F., Bowmani, R. and Minor, M.: Cats rule and dogs drool!: Classifying stance in online debate. In *Proceedings of the 2nd Workshop on Computational Approaches to Subjectivity and Sentiment Analysis*. pp. 1–9, Portland, OR, USA (2011).
- [15] Wojatzki, Michael, and Torsten Zesch: Stance detection in social media using stacked classifiers.” In *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, pp. 428–433. San Diego, California, (2016).
- [16] Lin, J., Kong, Q., Mao, W. and Wang, L.: A topic enhanced approach to detecting multiple standpoints in web texts. *Information Sciences*, pp.483–494, (2019).
- [17] Zhang, B., Yang, M., Li, X., Ye, Y., Xu, X. and Dai, K.: Enhancing cross-target stance detection with transferable semantic-emotion knowledge. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 3188–3197 (2020).
- [18] Liang, B., Fu, Y., Gui, L., Yang, M., Du, J., He, Y. and Xu, R.: Target-adaptive graph for cross-target stance detection. In *Proceedings of the Web Conference*, pp. 3453–3464 (2021)
- [19] Yan, Y., Chen, J. and Shyu, M.L.: Efficient large-scale stance detection in tweets. In *Deep Learning and Neural Networks: Concepts, Methodologies, Tools, and Applications*, pp. 667–683 (2020).
- [20] Wei, P., Lin, J. and Mao, W.: June. Multi-target stance detection via a dynamic memory-augmented network. In the proceedings of the 41st International Conference on ACM-SIGIR, pp. 1229–1232, (2018).
- [21] Liu, R., Lin, Z., Tan, Y. and Wang, W.: Enhancing zero-shot and few-shot stance detection with commonsense knowledge graph. In *Proceedings of the ACL-IJCNLP*, pp. 3152–3157, (2021).
- [22] Igarashi, Y., Komatsu, H., Kobayashi, S., Okazaki, N. and Inui, K.: Tohoku at SemEval-2016 task 6: Feature-based model versus convolutional neural network for stance detection. In *Proceedings of the 10th international workshop on semantic evaluation*, pp. 401–407, (2016).
- [23] Rajendran, G., Chitturi, B. and Poornachandran, P.: Stance-in-depth deep neural approach to stance classification. *Procedia computer science*, 132, pp.1646-1653 (2018)
- [24] Sobhani, P., Inkpen, D. and Zhu, X.: Exploring deep neural networks for multitarget stance detection. *Computational Intelligence*, 35(1), pp.82-97 (2019).
- [25] Sun, Q., Wang, Z., Li, S., Zhu, Q. and Zhou, G.: Stance detection via sentiment information and neural network model. *Frontiers of Computer Science*, 13, pp.127-138 (2019).
- [26] Hartmann, J., Heitmann, M., Siebert, C. and Schamp, C.: More than a feeling: Accuracy and application of sentiment analysis. *International Journal of Research in Marketing*, 40(1), pp.75–87 (2023)
- [27] Kenton, J.D.M.W.C. and Toutanova, L.K., 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT* (pp. 4171–4186).